

Walmart

You are a Data Analyst for Walmart....

Question 1: What is the total count of orders with reliable shipping (i.e., delivered on or before the expected delivery date) for each product category in December 2024?

```
SELECT f.product_category_id,  
       category_name, COUNT(*) reliable_order_count  
FROM fct_orders AS f  
INNER JOIN dim_product_categories AS d  
ON f.product_category_id = d.product_category_id  
WHERE actual_delivery_date <= expected_delivery_date  
       AND TO_CHAR(order_date, 'YYYY-MM') = '2024-12'  
GROUP BY f.product_category_id, category_name
```



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Question 2: Calculate the percentage of orders with unreliable shipping for each product category in December 2024. We define unreliable shipping as orders that were delivered after the expected delivery date, or orders that don't currently have an actual delivery date.

```
SELECT
    f.product_category_id,
    INITCAP(d.category_name) AS category_name,
    COUNT(*) AS total_order_count,
    COUNT(*) FILTER (
        WHERE actual_delivery_date > expected_delivery_date OR actual_delivery_date IS NULL
    ) AS unreliable_order_count,
    ROUND(
        COUNT(*) FILTER (
            WHERE actual_delivery_date > expected_delivery_date OR actual_delivery_date IS NULL
        )::DECIMAL / COUNT(*) * 100, 2
    ) AS unreliable_shipping_percentage
FROM fct_orders AS f
JOIN dim_product_categories AS d
    ON f.product_category_id = d.product_category_id
WHERE TO_CHAR(order_date, 'YYYY-MM') = '2024-12'
GROUP BY f.product_category_id, d.category_name;
```



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Question 3: Rank product categories into four performance quartiles based on the percentage of reliable shipping in December 2024. Identify which quartile each product category falls into. Recall that reliable shipping is defined as orders delivered on or before the expected delivery date.

```
with reliable AS(
    SELECT f.product_category_id,
           category_name, COUNT(*) AS reliable_order
    FROM fct_orders AS f
    INNER JOIN dim_product_categories AS d
    ON f.product_category_id = d.product_category_id
    WHERE actual_delivery_date <= expected_delivery_date
           AND TO_CHAR(order_date, 'YYYY-MM') = '2024-12'
    GROUP BY f.product_category_id, category_name),

total AS(
    SELECT f.product_category_id,
           category_name, COUNT(*) AS total_order
    FROM fct_orders AS f
    INNER JOIN dim_product_categories AS d
    ON f.product_category_id = d.product_category_id
    WHERE TO_CHAR(order_date, 'YYYY-MM') = '2024-12'
    GROUP BY f.product_category_id, category_name),

combined AS(
    SELECT
    t.product_category_id,
    t.category_name,
    t.total_order,
    COALESCE(r.reliable_order, 0) AS reliable_order,
    ROUND(1.0 * COALESCE(r.reliable_order, 0) / t.total_order *
100, 2) AS reliable_shipping_percentage
    FROM total t
    LEFT JOIN reliable r
        ON t.product_category_id = r.product_category_id)

SELECT
    NTILE(4) OVER (ORDER BY reliable_shipping_percentage DESC) AS quartile,
    category_name, reliable_shipping_percentage
FROM combined;
```

